

Assignment: Global Factor Data Analysis

5376 Asset/Risk Management I

Date due: June 07th, 2026 (23:59 P.M.)

Objective

In this assignment, students are tasked with analyzing global factor performance using factor returns data from the Global Factor Data repository. The analysis should include a structured approach to factor interpretation, performance assessment, and visualization, providing insights into how these factors behave in different market conditions.

Global Factor Performance Analysis

a.) Factor Selection: Choose between 2-3 factors from the [Global Factor Data](#) repository. Ensure that your choice of factors is unique from other groups in the class. Please fill out the factors sheet (**deadline: April 20th, 2026**) before you start working on the assignment so that others can see which combinations have already been chosen to avoid identical combinations for two groups. You can find the Factor selection Sheet [here](#) (Note: you must use your WU e-mail address).

b.) Factor Interpretation and Intuition: Briefly describe the intuition and economic interpretation behind your choice of factors. Additionally, explain whether your chosen factors are conceptually related, detailing any potential interactions or correlations. You should cite the original papers listed in the [Factor Data Documentation](#) and/or other sources.

c.) Historic Performance and Risk Analysis: Evaluate key risk-return characteristics of the selected factors, using rolling window analysis to examine performance variations over time. Specifically, assess historical performance before and after publication and compare against the market (U.S.; equally or value-weighted), benchmark indices and/or alternative asset classes.

d.) Spanning Regressions: Use the CAPM, the Fama and French (1993) three-factor model, and the Hou et al. (2021) five-factor model to assess factor exposures. Discuss the explanatory power of these models. How does their performance differ? Assess whether an alternative factor model might be more suitable.

Interpretation

Provide visualizations that support your analysis, such as cumulative return plots, rolling window statistics, and risk-return comparisons. Highlight any observed patterns or periodic breaks in performance. Interpret the results of your spanning regressions, discussing the extent to which your factors are explained by various existing factor models.

Further Instructions

- This is a **group assignment**. Students should independently form groups of two students and register themselves for a group on Canvas. **The deadline for this is April 20th, 2026!**
- Upload your group's solution by **June 07th, 23:59 P.M.**
- Your upload should be **one .zip file** consisting of three files:
 - **R-code:** Filename `<group_no>_code.R`, e.g., `group_2_code.R`
 - The **data** needed to run the code: `<group_no>_data.RData`
 - **Slides** for an 8-10 minute presentation (Note: this **includes** the Q&A) in `<group_no>_slides.pdf`
- Comment your R-code clearly. It must run without error once we open it from the folder where the data file is saved. If you use any packages, your code has to check if these packages are installed on the computer from which the code is run, and if not, auto-install missing packages from CRAN. **Please note: Packages not available from CRAN are not allowed.**
- The presentation should have a title page, 6–8 content slides, and one slide with the bibliography. Do not overload the slides with text, but be prepared to go into detail during the presentation. Every student must be able to present and answer questions on any part of the group assignment. The presentation will take place on **June 19th**. A list of assigned time slots will be posted on Canvas.

Literature

Fama, E. F., and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.

Hou, K., Mo, H., Xue, C., and Zhang, L. (2021). An augmented q-factor model with expected growth. *Review of Finance*, 25(1), 1–41.

Jensen, T. I., Kelly, B., and Pedersen, L. H. (2023). Is there a replication crisis in finance? *The Journal of Finance*, 78(5), 2465–2518.